

Project Almanac Example Document

Please review this document before continuing to task. The goal of this document is to highlight good and bad examples for this project in terms of browsing-requirements and contrived prompts.

Requirement 1: Your prompt MUST be browsing-heavy.

The context of your prompt should require SOME level of familiarity with the domain, but browsing is a non-negotiable part of the task. Prioritize browsing dependence over high specific domain reasoning.

✓ GOOD Example: browsing-heavy despite little domain reasoning

Domain: Biology
Prompt: In April 2024, the FDA approved the first targeted oral therapy for a rare autosomal dominant immunodeficiency defined by an acronym that spells a medical symptom. The phase 3 pivotal trial published in Blood in July 2024 was led by an Italian professor at the University of Brescia whose most-cited first publication, co-authored during his post-doctoral training at a US federal research institute from 1993 to 1995, identified a chemotactic role for a specific acute-phase protein. Find the senior corresponding author of that first publication. That scientist served as chief of a named laboratory within that same federal institute for over 30 years. Report only the full official name of that laboratory as it appeared in publications in 2024.
GTFA: Laboratory of Molecular Immunoregulation
Solution: Step 1- FDA approval + acronym=medical symptom Step 2- Phase 3 blood 2024 first author Raffaele Badolato, University of Brescia Step 3- His NCI postdoc and most cited first paper is Serum amyloid chemoattractant paper Step 4- Oppenheim's lab name at NCI in 2024 is Laboratory of Molecular Immunoregulation

✗ BAD Example: Domain reasoning, but lacks enough browsing

- The only browsing required is highlighted in yellow.

Domain: Physics
Prompt: Consider a 2x2 Bayer-mesh in a Photonic Integrated Circuit (PIC) composed of unbalanced Mach-Zehnder interferometers (MZIs). Select among the different technologies available for producing PIC, the technology with the largest market share as of March 2026. Each MZI block must be modeled as an unbalanced interferometer. The path-length difference of the

interferometers are chosen as a multiple of a base ΔL . This means that inside the mesh one MZI has an unbalance of ΔL , one of $2\Delta L$, one of $3\Delta L$ and the last one of $4\Delta L$. The shortest unbalance is chosen such that the resulting spectral response exhibits a periodicity of n full wavelength cycles across the wanted optical band used. The wavelength range to be considered is the O-band used in modern datacenters. $n=44$ is the number of channels available in that communication band. Compute the required imbalance ΔL needed to produce the n -cycle repetition within that band, assuming TE-polarized light and single-mode operation. Consider typical dimensions of the waveguide in the technology found.

Using these assumptions, determine the difference between the longest and shortest geometrical path connecting any input port to any output port in the 2x2 Bayer-mesh and express the answer in micrometers. Round the answer to a whole integer. Consider that the maximum difference between the longest and shortest interconnecting waveguides (connecting MZI inside the mesh) is of $50\text{ }\mu\text{m}$. Assume negligible propagation losses and a linear behavior of effective index as a function of wavelength.

GTFA:
1306 μm

Solution:

1. Identify Silicon Photonics as the technology with the largest market share as of March 2026.
2. Identify the extension of O band as 1260-1360 nm.
3. Identify the physical property of Silicon relevant for our case. Typical waveguide dimensions of 220 nm x 400 nm. TE group index in the middle of O band for this waveguide is approximately 4.2.
4. Determine the unbalance dL needed to achieve 44 FSR in the O band. dL is approximately 179.5 μm .
5. Identify the 2x2 Bayer mesh architecture as formed by two input and two output MZI.
6. Reason that to have the largest optical path possible, the input MZI should be the $3dL$ and the output the $4dL$ (or viceversa) with the longest interconnection (+50 μm in respect to other interconnection) between them.
7. Reason that to have the shortest optical path possible, the light travel in the arms without any dL and with the shortest interconnection.
8. Calculate the difference, round it to the nearest integer and expressed it in micrometer as 1306 μm .

Requirement 2: Your prompt must NOT be contrived

We have seen a number of highly contrived prompts. These are not allowed for this project, as they do not mirror realistic demands of conducting research online. If your request asks for variables that require a transformation with no applications of the results (adding indices of characters together, calculations with zero practical application), your prompt is likely too contrived. **Tasks will fail if not realistic.**

— BAD examples: Highly contrived prompts, using information in non-intuitive ways

- The highlighted portions showcase the contrivedness of the prompts

Prompt:

Identify the specific 1923 chemical engineering process for the atmospheric oxidation of ammonia that utilizes a specialized platinum-rhodium wire-gauge catalyst; take the first letter of this process's eponym and find its position in the English alphabet (A=1, B=2). Multiply this position by the specific operating pressure (in atmospheres) of the Lazote-Dupont plant in Belle, which is documented in the West

Virginia Encyclopedia. Divide the resulting product by a factor of 7.81, and use the nearest resulting integer (ignoring any decimals) as a four-digit year to identify the winner of the Perkin Medal. Name the only chemical process invented by the individual that bears the inventor's name.

Prompt:

Of the countries that were part of the United Nations on January 2, 1942, find the one whose capital on that date was the 12th closest to the North Pole compared to the capitals of the other countries. Next, get the GFP military index of this country as of the date that warlord Amade Oueremi was sentenced, and return the product of this number and the number of tanks of this country on that same date. Now, assume that this number represents Bitcoin, and calculate the USD value of that Bitcoin rounded to the nearest penny, according to the Yahoo Finance open price on March 10, 2020. Next, using the company that has a Spanish business conference on its economic calendar for March 30, 2026, get their historical high for RVN to BTC on March 31, 2019. Now take the previous USD value you had, add to it the latitude seconds of the country determined earlier, then multiply this by the latitude minutes, subtract the latitude degrees, and add the square root of the historical high from the previous sentence. Return just this number.

Using Judaism as the religion of interest within North America:

1. From Pew Research Center (2020), take the number of adherents based on self-identification.
 2. From Encyclopaedia Britannica, take the number of adherents based on active practice.
 3. Use a global population of exactly 8,000,000,000.
 4. Convert both values into percentages of the global population.
 5. Compute the absolute difference between the two percentages.
- Return only the numerical difference rounded to 5 decimal places.

It is VERY clear in some of these prompts that the writer is reverse engineering the prompt from an already established GTFA. A clear sign of reverse engineered prompts is giving highly specific information in the prompt.

- A resulting web-search would almost exactly mirror searching an exact paper/researcher/publication name, instead of browsing through multiple results to find a match.

Your goal should be to mirror a real research process, not to create a "research sounding process" from an end result.

❌ BAD Example: Reverse-engineered prompt

- Most of the information is included in the prompt, **the web-browsing is introduced in a contrived manner to "fill-in-the-blanks"**

A 2017 Cell Chemical Biology review from Yale looked at targeted protein degradation, specifically PROTACs. One molecule in particular was worth noting, and unlike most BET-targeting degraders at that time, it took out one BET bromodomain family member and largely left the other two alone. The same year, a group published the crystal structure of this PROTAC caught in the act, bound to both its target and its recruited E3 ligase at once. It was the first ternary complex crystal structure ever solved for a PROTAC, and the review cited the PDB accession code directly. That structure identifies the E3 ligase substrate recognition component. The small molecule ligand used to recruit that component was first described in a paper cited in the same review, where its dissociation constant came in at the nanomolar level. Report only the full name of the journal in which that original small-molecule ligand discovery paper was published.

Well Written Prompts:

✓ **GOOD example: browsing-heavy prompt that requires background knowledge in medicine**

Domain: Medicine
Prompt: Identify all automated upper-arm blood pressure monitors commercially available in both the EU and US between 2015-2024 that are validated according to ANSI/AAMI/ISO 81060-2:2018. From these, retain only devices that have at least one independent clinical validation study (defined as studies with no manufacturer funding or no conflict of interest). For each device, extract the reported mean systolic blood pressure error (in mmHg) from each eligible study and compute the unweighted average across studies. Exclude studies with sample sizes of < 85 or those using a non-standard measurement protocol. Among the remaining devices, determine which device has the lowest reported mean systolic difference. Report only the exact model name.
GTFA: JOYTECH DBP-1333b
Solution: Step 1 : Identify all automated upper-arm blood pressure monitors validated under ANSI/AAMI/ISO 81060-2:2018 between 2015–2024 using validated device listings and published studies. Step 2: Filter devices to include only those with sample size ≥ 85 and use of a standard validation protocol. Retain the eligible devices after filtering: OMRON 1. Omron 10 Series BP7450 (HEM-7342T-Z) 2. Omron Platinum BP5450 (HEM-7343T-Z) 3. Omron HBP-M4500 (clinical/NIBP module) 4. Omron HEM-1026 / HCR-1901T2 / HCR-1902T2 5. Omron BP5460 (HEM-7372T1-AZAZ) WITHINGS 6. Withings BPM Connect (WPM05) 7. Withings BPM Connect — Pregnancy/Pre-eclampsia validation 8. Withings BPM Core (WPM04) JOYTECH 14. JOYTECH DBP-1333b A study evaluated the accuracy of the JOYTECH DBP-1333b upper-arm blood pressure measuring device in the adult population according to the AAMI/ESH/ISO universal standard (ISO 81060-2:2018+Amd.1:2020). The study was approved by the ethics committee of the Affiliated Suzhou Science and Technology Town Hospital of Nanjing Medical University. Both criteria 1 and 2 were met. Stride BP Study: Hou G et al. Blood Press Monit 29(3):149–155, 2024. PMID: 38193368. PMC11045392 15. JOYTECH DBP-6279B NOTE: Most of the devices from Omron and withing were excluded due to the unavailability of

independent or no conflict of interest studies.

Step 4 : Extract the reported mean systolic BP difference (mmHg) for each device from its corresponding validation study:

Present as a stated fact in each of the validation studies included.

Step 5: Compare the extracted values across all eligible devices to determine the lowest mean systolic BP difference.

Step 6: Select the device with the minimum value.

JOYTECH DBP-1333b

✓GOOD example: first part of the prompt is browsing-heavy, final request is domain-specific

Domain:
Engineering

Prompt:

Among all of the molecular formulas for $C_4H_{10}O$ listed in the NIST Chemistry WebBook as of 2023, collect the molecules whose NIST-listed normal boiling point, NBP, as determined using a comparative ebulliometry extrapolation, is numerically closest to 375 K. Retrieve the vapor pressure of that compound at 350 K using the Antoine equation parameters obtained using a purification process that relied on comparative ebulliometry where temperature was in Kelvin and in the form of $A - B/(T+C)$. End your analysis by computing the standard Gibbs energy of vaporization at 350 K using $\Delta G_{\text{vap}} = RT \ln(P^o/P_{\text{vap}})$. Report the result in J/mol, rounded to the nearest integer.

GTFA:
\$2677 J/mol\$

Solution:

Step 1: Extract all $C_4H_{10}O$ isomers

From the NIST WebBook, the molecules that have that chemical formula are as follows:

1. 1-Butanol
2. 2-Butanol
3. 2-Methyl-1-propanol
4. 2-Methyl-2-propanol
5. Diethyl ether
6. Methyl propyl ether
7. Methyl isopropyl ether
8. Methyl isopropyl ether

Step 2: Determine the normal boiling point for each molecule, and find the closest one to 375 K. We must be careful here as the NIST has more than one boiling point for each molecule. We check the source for each, and determine that there are two methods used to determine the NBP:

The first method is using an experimental calorimeter procedure, while the second method relies on ebulliometry vapor pressure data extrapolation. The prompt explicitly says to rely on the ebulliometry data sets, so we'll use these NBPs.

Following the same number formatting shown above:

1. 390.9 K
2. 372.7 K
3. 381 K
4. 355.6 K
5. 3007.6 K
6. 307.6 K
7. 312 K
8. 303.9 K

The closest one is 2-Butanol, with a NBP at 372.66 K

Step 3: Obtain Antoine equation data for 2-butanol

The prompt explicitly says we need to use the Antoine parameters calculated using the purification process that relied on comparative ebulliometry. This method also lines up with the requested range of temperature that we have, as these parameters are obtained within a temperature range of 245 to 380 K. The other set of parameters (model B's error) is obtained within a range of 422 - 535 K, making them invalid.

A = 4.32943
B = 1158.672
C = -104.683

Step 4: Compute P_{vap} at 350 K using the Antoine equation:

$$\log_{10}(P) = 4.32943 - \frac{1158.672}{350 + -104.683}$$

$$P_{\text{vap}} = 10^{-0.39373} = 0.40389 \text{ bar}$$

Step 5: Verification

We can verify that at NBP (372.66 K), the same equation gives 1.013 bar. If we were to use the other equation, it would provide us with a different pressure from atmospheric, meaning that the other Antoine equation does not apply here under these conditions.

Step 6: Compute ΔG_{vap} at 350 K

$$\Delta G_{\text{vap}} = RT \ln\left(\frac{P^o}{P_{\text{vap}}}\right) = 8.314 \cdot 350 \text{ K} \cdot \ln\left(\frac{1.01325}{0.40389}\right) = 2677 \text{ J/mol}$$

✓ GOOD example: browsing-heavy

Domain:
Engineering

Prompt:

In European countries where 2024 motor vehicle production exceeded 1 million units, identify peer reviewed publicly accessible manufacturing or occupational case studies published in those countries between January 1, 2000 and December 31, 2024 that assess mental workload using NASA-TLX in automotive industry, and identify the largest sample size reported in any of the qualifying studies.

Report only the sample size.

GTFA:
250

Solution:

1. Identify all the European countries where 2024 motor vehicle production exceeded 1 million units:

-Germany: 4,069,222

-Spain: 2,376,504

-Czech Republic: 1,458,892

-Türkiye: 1,365,296

-France : 1,357,701

2. Define what NASA-TLX so that the correct search can be made:

NASA-TLX is a workload assessment tool used in human factors and ergonomics research to measure how demanding a task feels to a person. It typically measures workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration.

3. Define the literature search process to ensure that an exhaustive list is found and literature saturation is reached. The process to achieve this follows the usual research literature review process:

Within the identified countries, search for peer-reviewed studies published between January 1, 2000 and December 31, 2025 inside the country, using combinations of country names with terms: automotive, manufacturing, occupational, mental workload, NASA-TLX, and Task Load Index. At this stage it is important to clearly distinguish the difference between "published in" and "study conducted in". "Published in" refers to the country where the publisher operates in, while "study conducted in" refers to the location where the study was conducted in. The prompt explicitly asks for "published in".

Screen the retrieved records for relevance to automotive settings and confirm that NASA-TLX was explicitly used to assess mental workload.

Apply the inclusion criteria: publicly accessible study, published within the specified date range, conducted in one of the qualifying countries, situated in an automotive context, and using NASA-TLX as a workload assessment tool.

Apply the exclusion criteria: study outside the eligible countries or years, not publicly accessible, not in an automotive setting, not using NASA-TLX directly, or not an empirical case study.

Review the remaining eligible records in more detail to confirm final qualification and extract the reported sample size.

Use backward reference checking and keyword variations to identify any additional eligible studies that may not have appeared in the initial search. This is an iterative process that continues until saturation is reached. Saturation is defined as the point at which repeated searches across all qualifying countries and term combinations yield no new eligible studies and only return duplicates or clearly ineligible records.

Through the process mentioned above, qualifying articles include:

Turkey: Assessment of musculoskeletal disorders and mental workload among office workers of an automotive supplier company: 250

Germany: Field study of age-differentiated strain for assembly line workers in the automotive industry : 23

Germany: A full shift field study to evaluate user- and process-oriented effects of smart glasses on order picking in an automotive production plant: 23

Turkey: Evaluation of Mental Workload by Using Interval Type-2 Fuzzy-Based NASA-TLX Method: A Case Study in Automotive Industry: 40

4. Identify the study with the most participants among the qualifying studies:

Turkey: Assessment of musculoskeletal disorders and mental workload among office workers of an automotive supplier company: 250

Some additional clarifications:

Your final request can either be domain specific or related to your overall query:

- Domain Specific: a calculation result, property, method, etc.
- Tangentially Related: author's name, journal title, year of publication

You may construct calculation tasks that rely on browsing to fetch variables:

- Ex: "Calculate $X - Y$, where X is the experimentally measured weight according to journal articles published from 1990-1995, and Y is the theoretical weight according to journals published after the discovery of W ."